

5. RecyclerView – Student List App

Topics covered:

RecyclerView, Adapter, ViewHolder, LayoutManager, Toast

Task:

- Display a list of students (Name + Roll No.) using RecyclerView
 - On item click → show Toast with selected student name
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Step 1: Create New Project

1. Go to
File → New → New Project
 2. Select **Empty Views Activity**
 3. Click **Next**
-

Step 2: Project Details

1. Project Name:
StudentRecyclerView_Practical_5
 2. Language: **Kotlin**
 3. Minimum SDK: **API 21 or above**
 4. Click **Finish**
-
-

Step 3: Open `activity_main.xml`

1. Go to
app → src → main → res → layout → activity_main.xml
2. Click it
3. Select **Split** tab (Code + Design)
4. Replace existing XML with the following code:

```
5. <?xml version="1.0" encoding="utf-8"?>
    <androidx.recyclerview.widget.RecyclerView
        xmlns:android="http://schemas.android.com/apk/res/android"
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
```

➡ This layout contains only RecyclerView because it handles the entire list.

Step 4: Create Item Layout (`item_student.xml`)

1. Right-click on **layout** folder
2. Select **New** → **Layout Resource File**
3. File name: **item_student**
4. Root element: **ConstraintLayout**
5. Click **OK**

Replace the XML with:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <TextView
            android:id="@+id/txtName"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Student Name"
            android:textSize="18sp" />

        <TextView
            android:id="@+id/txtRoll"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Roll No" />

    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

➡ This layout represents **one row** in RecyclerView.

Step 5: Open `MainActivity.kt`

1. Go to
app → **src** → **main** → **java** → **com.example.studentrecyclerapp_practical_5** → **MainActivity.kt**
2. Open the file
3. Add the following imports at the top:

```
import android.os.Bundle
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.recyclerview.widget.LinearLayoutManager
import androidx.recyclerview.widget.RecyclerView
```

Step 6: Add RecyclerView Code

Replace the existing code with:

```
class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val recyclerView = findViewById<RecyclerView>(R.id.recyclerView)

        recyclerView.layoutManager = LinearLayoutManager(this)

        val students = listOf(
            "Amit - 101",
            "Sneha - 102",
            "Rahul - 103",
            "Pooja - 104"
        )

        recyclerView.adapter = SimpleAdapter(students)
    }

    // Adapter inside MainActivity
    inner class SimpleAdapter(private val studentList: List<String>) :
        RecyclerView.Adapter<SimpleAdapter.ViewHolder>() {

        inner class ViewHolder(view: View) : RecyclerView.ViewHolder(view)
        {
            val txtName: TextView = view.findViewById(R.id.txtName)
            val txtRoll: TextView = view.findViewById(R.id.txtRoll)
        }

        override fun onCreateViewHolder(parent: ViewGroup, viewType: Int):
            ViewHolder {
            val view = LayoutInflater.from(parent.context)
                .inflate(R.layout.item_student, parent, false)
            return ViewHolder(view)
        }

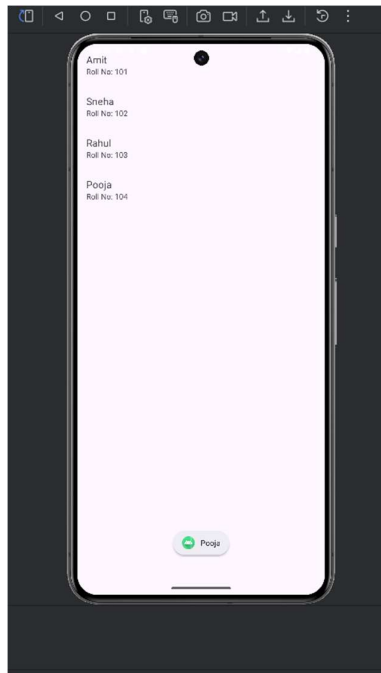
        override fun onBindViewHolder(holder: ViewHolder, position: Int) {
            val data = studentList[position].split(" - ")
            holder.txtName.text = data[0]
            holder.txtRoll.text = "Roll No: ${data[1]}"

            holder.itemView.setOnClickListener {
```

```
        Toast.makeText(  
            this@MainActivity,  
            data[0],  
            Toast.LENGTH_SHORT  
        ).show()  
    }  
}  
  
override fun getItemCount(): Int {  
    return studentList.size  
}  
}
```

Step 7: Run the App on Emulator

1. Click **RUN** 
2. App opens on emulator
3. Student list is displayed
4. Click on any item → Toast shows student name



Step 9: Connect App to Actual Phone (Same as Practical 2)

1. Open your phone → **Settings**
2. Go to **About device**

3. Tap Build Number 7 times
→ “You are now a developer!”
4. Go back → Additional Settings
5. Open Developer Options
6. Enable USB Debugging
7. Connect phone to laptop using USB cable
8. Allow USB debugging permission
9. Select your phone from device dropdown
10. Click RUN 

Output (Real phone):

